Trigonometric Functions and Circular Measure

ANSWER KEY

Section 1: Radian Measure, Arc Length and Sectors

1 A circle of radius 8 cm has a central sector with angle $\theta = \frac{3\pi}{4}$ rad. Find the exact arc length of the sector.

a. 6π cm ✓

b. 12π cm

c. 4π cm

d. 24π cm

Arc length $s = r\theta = 8\left(\frac{3\pi}{4}\right) = 6\pi$ cm.

2 Find the area of a sector of radius 6 cm having perimeter 22 cm.

a. 30 cm^2 ✓

b. 36 cm^2

c. 24 cm^2

d. 18 cm^2

Perimeter $= 2r + s = 12 + s = 22 \Rightarrow s = 10$ cm. Area $= \frac{1}{2}rs = \frac{1}{2}(6)(10) = 30$ cm^2.

3 A chord of length 10 cm subtends an angle of 2θ at the center of a circle of radius 13 cm. Find $\sin\theta$.

a. $\frac{5}{13}$ ✓

b. $\frac{12}{13}$

c. $\frac{10}{13}$

d. $\frac{5}{12}$

The chord is bisected: half-chord $= 5$ cm. In the right triangle, $\sin\theta = \frac{5}{13}$.

Section 2: Trigonometric Identities and Ratios

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4 Simplify the trigonometric expression $\frac{\sin\theta}{\cos\theta} + \frac{\cos\theta}{\sin\theta}$.

a. $\sec\theta\csc\theta$ ✓

b. $\tan\theta$

c. 1

d. $\sec^2\theta$

$\frac{\sin^2\theta+\cos^2\theta}{\sin\theta\cos\theta} = \frac{1}{\sin\theta\cos\theta} = \csc\theta\sec\theta$.

5 Given that $\tan\theta = -\frac{3}{4}$ and θ lies in the second quadrant, find the value of $\cos\theta$.

a. $-\frac{4}{5}$ ✓

b. $\frac{4}{5}$

c. $-\frac{3}{5}$

d. $\frac{3}{5}$

In quadrant 2, cosine is negative. With opposite 3 and adjacent 4, hypotenuse is 5: $\cos\theta = -4/5$.

6 Prove the identity $\sec^2\theta - \tan^2\theta = 1$. Which of the following equals $(\sec\theta - \tan\theta)(\sec\theta + \tan\theta)$?

a. 1 ✓

b. $\sec^2\theta$

c. $2\tan^2\theta$

d. 0

Difference of squares: $\sec^2\theta - \tan^2\theta = 1 + \tan^2\theta - \tan^2\theta = 1$.

Section 3: Solving Trigonometric Equations

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7 Solve $2\cos(2x) = 1$ for $0 \le x \le \pi$.

a. $x = \frac{\pi}{6}$ **and** $x = \frac{5\pi}{6}$ ✓

b. $x = \frac{\pi}{3}$ and $x = \frac{2\pi}{3}$

c. $x = \frac{\pi}{6}$ only

d. $x = \frac{\pi}{4}$ and $x = \frac{3\pi}{4}$

$\cos(2x) = 1/2 \Rightarrow 2x = \pi/3, 5\pi/3 \Rightarrow x = \pi/6, 5\pi/6$.

8 Find all solutions of $3\sin x = 2\cos^2 x$ in the interval $0^\circ \le x \le 360^\circ$.

a. 30° **and** 150° ✓

b. 30° and 210°

c. 60° and 120°

d. 45° and 135°

$3\sin x = 2(1 - \sin^2 x) \Rightarrow 2\sin^2 x + 3\sin x - 2 = 0 \Rightarrow (2\sin x - 1)(\sin x + 2) = 0 \Rightarrow \sin x = 1/2 \Rightarrow x = 30^\circ, 150^\circ$.

9 Solve the equation $\tan(x + 45^\circ) = 1$ for $0^\circ \le x \le 360^\circ$.

a. $x = 0^\circ, 180^\circ, 360^\circ$ ✓

b. $x = 45^\circ, 225^\circ$

c. $x = 90^\circ, 270^\circ$

d. $x = 0^\circ$ only

$x + 45^\circ = 45^\circ, 225^\circ, 405^\circ \Rightarrow x = 0^\circ, 180^\circ, 360^\circ$.

Section 4: Double Angle Formulae and Graph Amplitude

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10 If $\sin\theta = \frac{3}{5}$ and θ is acute, find the exact value of $\sin(2\theta)$.

a. $\frac{24}{25}$ ✓

b. $\frac{12}{25}$

c. $\frac{7}{25}$

d. $\frac{6}{5}$

$\cos\theta = 4/5$. $\sin(2\theta) = 2\sin\theta\cos\theta = 2(3/5)(4/5) = 24/25$.

11 State the amplitude and period of the function $y = 3\sin(4x) - 2$.

a. **Amplitude** $= 3$**, Period** $= \frac{\pi}{2}$ ✓

b. Amplitude $= 3$, Period $= \frac{\pi}{4}$

c. Amplitude $= 1$, Period $= 2\pi$

d. Amplitude $= 4$, Period $= \frac{\pi}{2}$

Amplitude is $|a| = 3$. Period is $2\pi/b = 2\pi/4 = \pi/2$.

12 Express $\cos(2\theta)$ purely in terms of $\sin\theta$.

a. $1 - 2\sin^2\theta$ ✓

b. $2\sin^2\theta - 1$

c. $1 - \sin^2\theta$

d. $\sin^2\theta - 1$

$\cos(2\theta) = \cos^2\theta - \sin^2\theta = (1 - \sin^2\theta) - \sin^2\theta = 1 - 2\sin^2\theta$.

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